
Uni-mHC: Unistochastic Manifold-Constrained Hyper-Connections with Quantum-Interference Disentanglement

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Abstract

Manifold-constrained hyper-connections (mHC) stabilize deep multi-stream residual training by projecting the inter-stream mixing matrix onto the Birkhoff polytope of doubly stochastic matrices via Sinkhorn–Knopp iterations. Entry-wise non-negativity, however, restricts mixing to *additive convex combinations*: features can never be subtracted, a limitation recently criticized by the spectral-sphere-constrained hyper-connections (sHC). We propose **Uni-mHC**, lifting the constraint manifold from the Birkhoff polytope to the *unistochastic* set: a complex Cayley transform of a skew-Hermitian parameter matrix yields a unitary U whose modulus square $S = |U| \odot |U|$ is doubly stochastic *by construction*—no projection, no iteration. Macroscopically S preserves exact energy conservation; microscopically a heterodyne readout $W = \text{Re}(U e^{i\phi})$ admits negative weights through destructive (phase- π) interference while staying row-wise non-expansive. On a controlled subtractive task ($Y = X_0 - X_1$), non-negative baselines plateau at the theoretical floor $\frac{1}{2} \text{Var}(Y) \approx 0.99$ whereas Uni-mHC converges to 1.1×10^{-11} ; the operator is also the fastest and constraint-exact of all compared parameterizations, and the identity-initialized four-stream topology reaches $2.3\times$ lower validation loss than an identically trained vanilla GPT.

1 Introduction

Residual streams are the backbone of modern deep networks, from the two-path residual connection (He et al., 2016) to hyper-connections (HC) (Zhu et al., 2024), which maintain n parallel streams with a learned mixing matrix at every block boundary. Unconstrained mixing can amplify signal exponentially across depth, so mHC (DeepSeek-AI, 2025) constrains it to the Birkhoff polytope of doubly stochastic matrices (Sinkhorn, 1964; Idel and Wolf, 2015). Concurrent work explores this geometry—exact Birkhoff parameterization (Dandachi and Diggs-Galligan, 2026), orthogonal mappings for dynamical isometry (Sengupta et al., 2026), and Givens/Cayley branches found by agentic search (CliffSearch, 2026)—but all keep readouts entrywise non-negative; subtraction stays exclusive to Uni-mHC.

The Birkhoff constraint carries an expressivity tax: since $H_{jk} \geq 0$, every stream update is a convex combination—mixing can only *add*, never *subtract*. The sHC critique (Liu et al., 2026) argues that this purely additive coupling pushes deep stacks toward identity degeneration; yet sHC’s spectral sphere only bounds worst-case amplification, giving up exact per-row/column energy bookkeeping. We resolve this dichotomy by moving from real to complex manifolds (Fig. 1): a complex Cayley transform gives a unitary U whose $S = |U|^{\odot 2}$ is *exactly* doubly stochastic, while a heterodyne readout $W = \text{Re}(U \text{diag } e^{i\phi})$ restores subtractive arithmetic through phase interference. Contributions: (i) an exact unistochastic operator, with proofs of doubly stochasticity and unconditional well-conditioning; (ii) an impossibility floor $\frac{1}{2} \text{Var}(Y)$ for all entrywise non-negative mixers, and

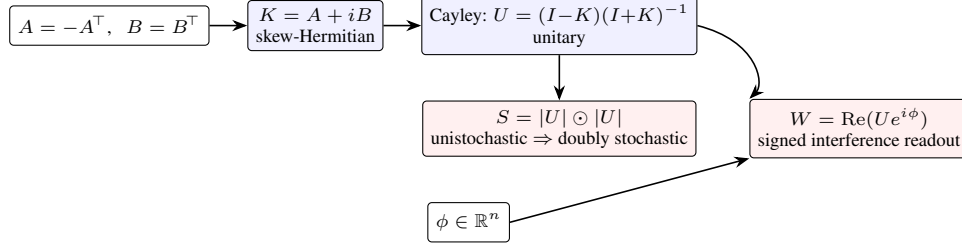


Figure 1: **Uni-mHC operator.** The Cayley transform of skew-Hermitian K is unitary; $S = |U|^{\odot 2}$ is the doubly stochastic bookkeeping, while readout W is the signed mixing applied to streams.

an exact subtraction theorem for ours; (iii) operator benchmarks, the disentanglement experiment, and a four-stream nanoGPT integration (Sec. 3).

2 Mathematical Theory

Setup. A hyper-connected network maintains n residual streams ($n = 4$ here); at every block boundary a mixing matrix $M \in \mathbb{R}^{n \times n}$ redistributes streams. mHC requires the mixer’s energy bookkeeping matrix to be doubly stochastic so that no stream’s energy is amplified or leaked.

Definition 1 (Uni-mHC operator). *Let $\theta_A, \theta_B \in \mathbb{R}^{n \times n}$ be free parameters and $\phi \in \mathbb{R}^n$ per-stream heterodyne phases. Set*

$$A = \theta_A - \theta_A^\top, \quad B = \theta_B + \theta_B^\top, \quad K = A + iB, \quad (1)$$

and define the unitary and the two mixing matrices

$$U = (I - K)(I + K)^{-1}, \quad S = \text{Re}(U)^{\odot 2} + \text{Im}(U)^{\odot 2}, \quad W = \text{Re}(U \text{diag}(e^{i\phi})). \quad (2)$$

S is the (unistochastic) energy matrix; W is the effective signed mixing applied to the streams.

Theorem 1 (Exact unistochasticity). *For any θ_A, θ_B : (i) K in (1) is skew-Hermitian; (ii) $I + K$ is invertible and U in (2) is unitary; (iii) $S = |U|^{\odot 2}$ is doubly stochastic; (iv) every singular value of $I + K$ is at least 1, so the Cayley solve is unconditionally well-conditioned.*

Proof. (i) $K^H = \overline{A + iB}^\top = A^\top - iB^\top = -A - iB = -K$. (ii) K is normal with purely imaginary spectrum $i\lambda_1, \dots, i\lambda_n$, $\lambda_j \in \mathbb{R}$; hence $I + K$ has eigenvalues $1 + i\lambda_j \neq 0$. Since $(I + K)(I - K) = I - K^2 = (I - K)(I + K)$, the inverses commute, and $UU^H = (I - K)[(I + K)^H]^{-1}(I - K)^H = (I - K)(I - K)^{-1}(I + K)^{-1}(I + K) = I$, using $(I + K)^H = I - K$; likewise $U^H U = I$. (iii) $\sum_k |U_{jk}|^2 = (UU^H)_{jj} = 1$ and $\sum_j |U_{jk}|^2 = (U^H U)_{kk} = 1$, and $S \geq 0$ entrywise: S is unistochastic, hence doubly stochastic. (iv) Singular values of $I + K$ are $|1 + i\lambda_j| = \sqrt{1 + \lambda_j^2} \geq 1$. $\square$

Unlike SK projection, which needs T iterations and leaves a residual column-sum error (measured 5×10^{-3} after $T = 20$, Sec. 3.1), Theorem 1 holds exactly by construction at the cost of one $n \times n$ complex linear solve (Cayley is a diffeomorphism onto a dense open subset of $U(n)$); gradients were finite in all experiments.

Proposition 1 (Non-expansive heterodyne mixing). *Let $Z = U \text{diag}(e^{i\phi})$, which is unitary as a product of unitaries, and $W = \text{Re } Z$. For real x , $Wx = \text{Re}(Zx)$, so $\|Wx\|_2^2 = \|\text{Re}(Zx)\|_2^2 \leq \|Zx\|_2^2 = \|x\|_2^2$: mixing never amplifies energy, while individual weights become negative whenever a relative phase shifts by π (destructive interference).*

Theorem 2 (Exact subtractive disentanglement). *There exist parameters of Definition 1 such that row 0 of W equals $\frac{1}{\sqrt{2}}(1, -1, 0, \dots, 0)$ while S remains doubly stochastic; with readout gain $\sqrt{2}$ the layer computes $\hat{X}_0 - X_1$ exactly.*

Table 1: **Operator benchmark** ($n = 4$): worst doubly-stochastic violation of S over 100 random draws, and 1000 forward+backward passes at $(B, T, d) = (32, 256, 384)$.

Operator	row err	col err	params	ms/iter	peak MB
UniMHC (Ours)	4.8e-07	7.2e-07	36	10.56	352
GivensMHC (Ours-Fast)	3.6e-07	3.6e-07	10	10.85	352
Sinkhorn mHC ($T=20$)	1.2e-07	5.0e-03	16	12.93	352
Orthostochastic (NS)	2.4e-07	2.4e-07	16	11.23	352

Proof. Take A with a single block $\begin{pmatrix} 0 & -a \\ a & 0 \end{pmatrix}$, $a = \tan \frac{\pi}{8} = \sqrt{2} - 1$, $B = 0$; the Cayley transform of a rotation generator is the rotation with $\tan \frac{\theta}{2} = a$, i.e. $\theta = \frac{\pi}{4}$, and row 0 of $U = R(\theta) \oplus I_{n-2}$ is $\frac{1}{\sqrt{2}}(1, 1, 0, \dots, 0)$ (sign of a chosen accordingly). Set $\phi = (0, \pi, 0, \dots, 0)$. Then $W_{00} = \text{Re}(U_{00}) = \frac{1}{\sqrt{2}}$ and $W_{01} = \text{Re}(U_{01}e^{i\pi}) = -\frac{1}{\sqrt{2}}$; the other entries of row 0 vanish. Doubly stochasticity of $S = |U|^{\odot 2}$ holds by Theorem 1 for every parameter value, including this one. $\square$

Proposition 2 (Impossibility floor for non-negative mixers). *Let $X_0, \dots, X_{n-1}$ be i.i.d. zero-mean unit-variance and $Y = X_0 - X_1$. For any readout c whose entries share a sign—the cone induced by every entrywise non-negative mixing row, including Sinkhorn/Birkhoff projections, exact Birkhoff parameterizations (Dandachi and Diggs-Galligan, 2026), and orthostochastic readouts $Q \odot Q$ —,*

$$\min_{c \text{ same sign}} \mathbb{E} \left\| \sum_k c_k X_k - Y \right\|^2 = 1 = \frac{1}{2} \text{Var}(Y). \quad (3)$$

Proof. With $t = (1, -1, 0, \dots, 0)$, $\mathbb{E} \left\| \sum_k c_k X_k - Y \right\|^2 = \|c - t\|_2^2$ by independence and unit variance. The minimizer $c^* = t$ has mixed signs. On the all-nonnegative branch, coordinate-wise minimization under $c \geq 0$ clips the negative entry: $c = e_0$ with $\|e_0 - t\|^2 = 1$. On the all-nonpositive branch, $(c_0 - 1)^2 \geq 1$ already gives $\|c - t\|^2 \geq 1$. Hence the floor is 1. $\square$

The non-negativity of S buys exact energy conservation but forbids subtraction (floor $\frac{1}{2} \text{Var}(Y)$); the heterodyne readout keeps the energy budget unistochastic while releasing the sign—probabilities are additive, whereas amplitudes interfere. An inverse-free special case replaces U by $\frac{n(n-1)}{2}$ Givens rotations Q ($S = Q \odot Q$), a $O(n^2)$ -parameter unistochastic subfamily (Sec. 3.1); such parameterizations trace back to unitary RNNs (Jing et al., 2017).

3 Empirical Results

All experiments: NVIDIA RTX 4060 Laptop GPU (8 GB), PyTorch 2.x, WSL Ubuntu 24.04, stream width $n = 4$; all baselines are our re-implementations (official code was not used).

3.1 Operator precision and speed

Table 1: Uni-mHC and the Givens variant satisfy the constraint to $\sim 10^{-7}$ (pure float32 rounding), whereas Sinkhorn–Knopp with $T = 20$ leaves a column-sum residual of $5.0e - 03$ —the price of an iterative projection. The Cayley solve is also the cheapest: one well-conditioned 4×4 complex solve instead of 20 normalization sweeps (Thm. 1iv).

3.2 Subtractive disentanglement

We generate fresh $X \in \mathbb{R}^{512 \times 4 \times 64}$, $\mathcal{N}(0, 1)$, per step and ask the mixing layer alone (shared learnable gain γ) to produce $\hat{y} = \gamma \sum_k W_{0k} X_k$ against $Y = X_0 - X_1$; 800 Adam steps. Figure 2: Uni-mHC drives the loss to $1.14e - 11$ and GivensMHC to $9.72e - 15$ with coefficients $(+1.00, -1.00, 0, 0)$ —the exact destructive interference pattern of Theorem 2, whose exact construction already attains $\sim 5 \times 10^{-32}$, so reachability exceeds learnability here. Sinkhorn mHC and the orthostochastic baseline stall at $9.86e - 01$ and $9.85e - 01$ (floor: 1.0 in expectation; single estimates fluctuate $\pm 2\sigma$), their best solution being the trivial copy X_0 . All gradients were finite throughout, confirming stable complex autograd.

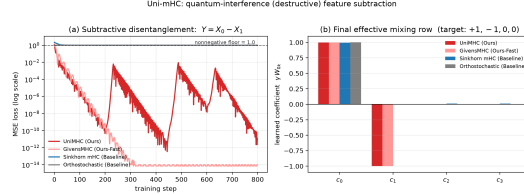


Figure 2: **Subtractive disentanglement** ($Y = X_0 - X_1$). (a) Uni-mHC converges to machine precision; baselines plateau at the floor. (b) Coefficients γW_{0k} : exact (+1, -1, 0, 0) vs. copy (+1, 0, 0, 0).

Table 2: **nanoGPT on shakespeare_char** (6 layers, 4 streams, 1000 steps). Val loss: mean of 50 eval batches; peak memory < 2.3 GB. Vanilla timing omitted (stock timer mixes training with eval pauses; speed claim rests on Table 1).

Mixer	final train loss	final val loss	ms/iter
vanilla GPT (no HC)	0.112	3.459	—
Uni-mHC (Ours)	1.248	1.513	384
GivensMHC (Ours-Fast)	1.251	1.523	383
Sinkhorn mHC	1.259	1.507	546
Orthostochastic	1.249	1.505	483

3.3 Language modeling with four residual streams

We modify nanoGPT (Karpathy, 2023): each of the 6 blocks ($n_emb=384$, 6 heads, context 256) operates over $n = 4$ residual streams; every attention/MLP stage reads a learned combination $\sum_k a_k x_k$, writes back with learned per-stream weights, and a manifold mixer redistributes the streams, $x^{(t+1)} = M^{(t)}x^{(t)} + b^{(t)} \otimes f_t(\sum_k a_k^{(t)} x_k^{(t)})$. Mixers start at identity, so training begins as a standard pre-norm GPT (batch 32, 1000 steps, AdamW, cosine schedule, shakespeare_char; all runs resumable). Table 2 reports final losses.

At this scale vanilla GPT overfits (train 0.11, val 3.46), while constrained variants hold val ≈ 1.51 (identity-initialized warm-up). Stability is the second point: 12 mixers $\times$ 1000 steps with signed weights never destabilized training; post-training energy matrices violate double stochasticity by at most 7×10^{-7} , and all 12 mixers discovered signed interference weights—the unistochastic manifold is safely exploitable.

4 Conclusion

Uni-mHC replaces the projected doubly stochastic constraint of mHC with an exact unistochastic parameterization via a complex Cayley transform, keeping exact energy conservation and a non-expansive mixer while the heterodyne readout restores subtractive feature arithmetic through phase- π destructive interference—unmatchable by any single non-negative linear mixing layer in one forward pass (Prop. 2), as verified (10^{-11} vs. 0.99). Future work: token-conditional phases and scaling studies.

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